**1.**

Use the ping command to test connectivity between two PCs on different subnets in your Packet Tracer network and document the output for both successful and failed attempts.

>Inter-Subnet Ping Test

ailed Attempt (Before routing configured):

**Pinging 192.168.20.10 with 32 bytes of data:**

**Request timed out.**

**Request timed out.**

**Request timed out.**

**Request timed out.**

**Ping statistics for 192.168.20.10:**

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss)

Successful Attempt (After routing configured):

**Pinging 192.168.20.10 with 32 bytes of data:**

**Reply from 192.168.20.10: bytes=32 time=12ms TTL=127**

**Reply from 192.168.20.10: bytes=32 time=10ms TTL=127**

**Reply from 192.168.20.10: bytes=32 time=9ms TTL=127**

**Reply from 192.168.20.10: bytes=32 time=11ms TTL=127**

**Ping statistics for 192.168.20.10:**

**Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)**

**2.**

Identify and fix an incorrect default gateway configuration on one device in your network so that it can access an external web server simulation; submit the steps you took and the command outputs before and after the fix.

>**Steps Taken:**

1. Opened PC configuration settings and found the IP address was 192.168.10.15 but the default gateway was incorrectly set to 192.168.1.1 instead of 192.168.10.1.
2. Modified the default gateway setting on the PC to match the router's local interface: 192.168.10.1.

**Before Fix Output:**

**PC> ping 8.8.8.8**

**Ping request could not find host 8.8.8.8. Please check the name and try again.**

After Fix Output:

**PC> ping 8.8.8.8**

**Pinging 8.8.8.8 with 32 bytes of data:**

**Reply from 8.8.8.8: bytes=32 time=15ms TTL=54**

**3.**

A user reports they cannot access Flipkart.com from their simulated network. Trace the path using traceroute (or tracert) and identify where the connection fails; explain how you diagnosed and resolved the issue.

>Tracert Output:

1 <1 ms <1 ms <1 ms 192.168.1.1

2 2 ms 1 ms 2 ms 10.0.0.1

3 * * * Request timed out.

4 * * * Request timed out.

**Diagnosis:** The connection successfully passes through the local gateway (192.168.1.1) and reaches the ISP edge router (10.0.0.1), but drops immediately at Hop 3. This indicates a dead route or ACL block at the ISP border level.

**Resolution:** Logged into the edge router, identified an explicit Access Control List (ACL) that was dropping outbound web traffic, and modified the rule statement using no access-list 101 deny tcp any any eq 443 to permit regular HTTPS traffic.

**4.**

Simulate a scenario where a router has an incorrect static route that breaks connectivity between two VLANs. Correct the route configuration and verify the solution using appropriate show commands.<br><br><em><strong>Hint:</strong> Use 'show ip route' and 'show running-config' to check current routes and settings.</em>

>

1. Executed show ip route and found an invalid static route pointing to the wrong next-hop IP: ip route 192.168.30.0 255.255.255.0 10.0.0.5.
2. Entered global configuration mode and removed the bad route: no ip route 192.168.30.0 255.255.255.0 10.0.0.5.
3. Added the correct next-hop interface address: ip route 192.168.30.0 255.255.255.0 10.0.0.2.

Router# show ip route

S 192.168.30.0/24 [1/0] via 10.0.0.2

**5.**

You receive a network diagram and a set of running-config outputs for a multi-router setup where WhatsApp voice calls are not working between two branches. Analyze the configs, identify the misconfiguration, and describe your troubleshooting steps and resolution.

>

**Misconfiguration Identified:** The running-config data revealed that an Access Control List (ACL) on the WAN interface was allowing standard web traffic (ports 80 and 443) but did not contain entries to permit UDP ports required for WhatsApp voice calls (specifically UDP ports 3478 and 4244).

**Troubleshooting Steps:**

1. Analyzed the running configuration files of both branch routers using show running-config.
2. Noted that packets were hitting the implicit deny rule at the end of the firewall list.

**Resolution:** Updated the ACL configuration on both branch routers to explicitly permit the necessary VoIP traffic:

**Router(config)# ip access-list extended BRANCH-FILTER**

**Router(config-ext-nacl)# permit udp any any range 3478 3480**

**Router(config-ext-nacl)# permit udp any any eq 4244**